Lab Report - 02

Experiment on 02

Course Name: Structured Programming Lab

Course Code: CSE-122

Submitted to

Rishad Amin Pulok

Senior Lecturer

Dept. of Computer Science & Engineering

Metropolitan University, Sylhet

Submitted by

Mahjabin Jahan Anika

ID: 251-115-083

Batch: 62

Section: B

Dept. of Computer Science & Engineering

Metropolitan University, Sylhet

Submission Date: 2025-06-24

Lab Report-02

INDEX

Serial No.	Question Name
01	Write a C program that accepts two courses' grades and credit hours of those courses (floating point values) and calculates your CGPA.
02	Write a C program that accepts an employee's ID, total worked hours in a month and the amount he received per hour. Print the ID and salary (with two decimal places) of the employee for a particular month.
03	Write a C program to calculate a bike’s average consumption from the given total distance (integer value) traveled (in km) and spent fuel (in litters, float number– 2 decimal points).
04	Write a C program to calculate the distance between two points.
05	Write a C program to print the roots of Bhaskara’s formula from the given three floating numbers.
06	Print the following pattern as it is.

Signature :

Date : 24/06/2025

Problem-1: Write a C program that accepts two courses' grades and credit hours of those courses (floating point values) and calculates your CGPA.

Test Data :

Grade in course-1: 3.75

Credit hour of course-1: 3

Grade in course-2: 3.50

Credit hour of course-2: 1.5

Expected Output:

CGPA =3.66

Input:

untitled 15.c

```c
#include<stdio.h>
int main()
{
    float cgpa,grade1,grade2,credit1,credit2;
    printf ("Grade in course-1:");
    scanf("%f",&grade1);
    printf("\nCredit hour of course-1:");
    scanf("%f",&credit1);
    printf("\nGrade in course-2:");
    scanf("%f",&grade2);
    printf("\nCredit hour of course-2:");
    scanf("%f",&credit2);

    cgpa=(grade1*credit1+grade2*credit2)/(credit1+credit2);
    printf("\nCGPA=%.2f\n",cgpa);
    return 0;
}
```

Output:

```
Grade in course-1:3.75

Credit hour of course-1:3

Grade in course-2:3.50

Credit hour of course-2:1.5

CGPA=3.67

Process returned 0 (0x0)   execution time : 18.108 s
Press any key to continue.
```

Problem-2:Write a C program that accepts an employee's ID, total worked hours in a month and the amount he received per hour. Print the ID and salary (with two decimal places) of the employee for a particular month.

Test Data :

Input the Employees ID: 0342

Input the working hrs: 8

Salary amount/hr: 15000

Expected Output:

Employees ID = 0342

Salary = 120000.00 BDT

Input:

```c
#include<stdio.h>
int main()
{
 float  wh,id,salary,rate;

 printf("Input the Employees ID:");
  scanf("%f",&id);
  printf("\nInput the working hrs:");
  scanf("%f",&wh);
  printf("\nSalary amount/hr:");
  scanf("%f",&rate );

  printf("\nEmployees ID=");
  scanf("%f",&id);
  salary=(wh*rate);
  printf("\nSalary=%.2f BDT\n",salary);

  return 0;
}
```

Output:

```
Input the Employees ID:0342

Input the working hrs:8

Salary amount/hr:15000

Employees ID=0342

Salary=120000.00 BDT

Process returned 0 (0x0)   execution time : 18.711 s
Press any key to continue.
```

Problem-3: Write a C program to calculate a bike's average consumption from the given total distance (integer value) traveled (in km) and spent fuel (in litters, float number– 2 decimal points).

Test Data :

Input total distance in km: 350

Input total fuel spent in liters: 5

Expected Output:

Average consumption (km/lt) 70.000

Input:

untitled 15.c

```c
#include<stdio.h>
int main()
{
    int distance;
    float fuel,average;
    printf("Input total distance in km:");
    scanf("%d",&distance);
    printf("Input total fuel spent in liters:");
    scanf("%f",&fuel);

    average=distance/fuel;

    printf("\nAverage consumption (km/lt)%.3f\n",average);

    return 0;
}
```

Output:

```
Input total distance in km:350
Input total fuel spent in liters:5

Average consumption (km/lt)70.000

Process returned 0 (0x0)   execution time : 4.551 s
Press any key to continue.
```

Problem-4: Write a C program to calculate the distance between two points.

Test Data :

Input x1: 25

Input y1: 15

Input x2: 35

Input y2: 10

Expected Output:

Distance between the said points: 11.1803

Input:

```c
#include<stdio.h>
#include<math.h>

int main()
{
    float x1,y1,x2,y2,distance;

    printf("Input x1:");
    scanf("%f",&x1);
    printf("Input y1:");
    scanf("%f",&y1);
    printf("Input x2:");
    scanf("%f",&x2);
    printf("Input y2:");
    scanf("%f",&y2);

    distance=sqrt(pow(x1-x2,2)+pow(y1-y2,2));
    printf("\nDistance between the said points:%.4f\n",distance);

   return 0;
}
```

Output:

```
Input x1:25
Input y1:15
Input x2:35
Input y2:10

Distance between the said points:11.1803

Process returned 0 (0x0)   execution time : 11.051 s
Press any key to continue.
```

Problem5: Write a C program to print the roots of Bhaskara's formula from the given three floating numbers.

Test Data :

Input the first number(a): 25

Input the second number(b): 35

Input the third number(c): 12

Expected Output:

Root1 =-0.60000

Root2 =-0.80000

Input:

```c
#include<stdio.h>
#include<math.h>
int main()
{
    float a,b,c,discriminant,root1,root2;

    printf("Input the first number(a):");
    scanf("%f",&a);
    printf("Input the secend number(b): ");
    scanf("%f",&b);
    printf("Input the third number(c):");
    scanf("%f",&c );

    discriminant=b*b-4*a*c;
    if (discriminant >=0)
      {
      root1=(-b+sqrt(discriminant))/(2*a);
      root2=(-b-sqrt(discriminant))/(2*a);

      printf("\nRoot1= %.5f\n",root1);
      printf("Root2= %.5f\n",root2);
     }
    return 0;
}
```

Output:

```
Input the first number(a):25
Input the secend number(b): 35
Input the third number(c):12

Root1= -0.60000
Root2= -0.80000

Process returned 0 (0x0)   execution time : 14.309 s
Press any key to continue.
```

Problem-6: Print the following pattern as it is.

Expected Output:

```
        *
      * * *
    * * * * *
  * * * * * * *
* * * * * * * * *
```

Input:

```c
#include <stdio.h>

int main() {
    printf("    *\n");
    printf("   ***\n");
    printf("  *****\n");
    printf(" *******\n");
    printf("*********\n");
    return 0;
}

```

Output:

```
    *
   ***
  *****
 *******
*********

Process returned 0 (0x0)   execution time : 0.137 s
Press any key to continue.
```